

Md Omar Faruqe Riyad

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Research interests: AI agents for site reliability engineering and software engineering; NLP for low-resource languages.

EDUCATION

Shahjalal University of Science and Technology (SUST)

Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering; CGPA: 3.53/4.00 (Last two years: 3.83)

2018 – 2023

- GRE: 317/340 · TOEFL: 104/120

RESEARCH EXPERIENCE

AI for Site Reliability Engineering (SREGym) | *Research Intern, advised by Prof. Tianyin Xu* UIUC, 2026

- Working on the SREGym v2 paper: extending [SREGym](#), an open-source benchmark for AI SRE agents, from synthetic faults to real-world failure scenarios, and analyzing how agents behave on them.
- Turning documented production outages into reproducible benchmark problems through case studies and postmortem analysis, then designing fault scenarios and oracles that target known agent weaknesses: cross-layer reasoning gaps, greedy diagnosis anchoring, and reward hacking.

Code Generation for Bangla | *BLP-2025 Shared Task 2, first author* 2025

- Fine-tuned Llama 3, TigerLLM, and Qwen with LoRA adapters for Bangla-to-Python code generation, a setting with little supervised data and no strong open baseline.
- Added Chain-of-Thought decomposition and a self-refinement loop that repairs generated code from execution feedback, reaching **Pass@1 0.85**; categorized failures by type (syntax, runtime, misread problem statement) to steer refinement.

Violence Inciting Text Detection in Bangla | *BLP-2023 Shared Task 1, first author* 2023

- Countered class imbalance and data scarcity with semi-supervised self-training over unlabeled Bangla text plus back-translation augmentation, then ensembled fine-tuned Bangla transformers with bagging and majority voting to cut variance on minority classes.

Visual Question Answering (VQA) Data Synthesis | *Undergraduate Thesis* 2022

- Augmented VQA v2 training data with automatically generated question-answer pairs to widen answer-space diversity, then measured where models fall back on answer-frequency bias instead of visual reasoning.

PUBLICATIONS

- [1] **Omar Faruqe Riyad** and Jahedul Alam Junaed. [AdversaryAI at BLP-2025 Task 2: A Think, Refine, and Generate \(TriGen\) System with LoRA and Self-Refinement for Code Generation](#). BLP Workshop at IJCNLP-AACL 2025.
- [2] Trina Chakraborty, Marufur Rahman, and **Omar Riyad**. [Deja Vu at SemEval 2024 Task 9: A Comparative Study of Advanced Language Models for Commonsense Reasoning](#). SemEval Workshop at NAACL 2024.
- [3] **Omar Faruqe Riyad**, Trina Chakraborty, and Abhishek Dey. [Team_Syrax at BLP-2023 Task 1: Data Augmentation and Ensemble Based Approach for Violence Inciting Text Detection in Bangla](#). BLP Workshop at EMNLP 2023.

EXPERIENCE

Dynamic Solution Innovators

April 2023 – Present

Software Engineer

Dhaka, Bangladesh

- Contributed to the data pipeline behind the voter registration flow of EC-BVRS, Bangladesh's national voter registration (NID) platform, which processes records for nearly **150 million voters** (Spring Boot, Kafka, Kudu, Redis).
- Resolved a production incident blocking payment processing after the Kafka pipeline fell **500K events behind**; traced recurring consumer rebalances to slow Kudu writes one layer below and reduced batch size to restore consumer liveness and clear the backlog.
- Built and open-sourced [gf](#) , a GlassFish CLI that hot-swaps modified bytecode into running JVMs over JDI/JDWP and bypasses JasperReports classloader caching. Cut local edit-deploy cycles from **~2 minutes to ~5 seconds**.
- Introduced Prometheus, Loki, and Grafana observability for GlassFish applications, exposing JVM memory, thread pools, and request latency for production debugging.

Altri.ai

Part-time

Full-Stack AI Engineer, RevestAI 

Remote, US

- Designed the three-stage property scoring pipeline that separates LLM feature extraction, ML valuation, and deterministic financial heuristics, so each layer can be retrained and evaluated on its own.
- Rewrote the sequential LLM enrichment pipeline as bounded-concurrency async batches with atomic per-batch commits. Throughput rose about **10×**, and failed runs resume without repeating completed LLM calls.
- Implemented distributed job coordination on PostgreSQL with **SELECT FOR UPDATE** row locks and stale-lock recovery, so multiple scheduler replicas safely share long-running enrichment and training jobs.

- Built the property ingestion pipeline on PostgreSQL with content-hash deduplication and incremental delta-syncing, so repeated scrapes skip already-enriched listings while preserving historical state. Model promotion is gated the same way: a challenger must beat the champion on a gold-standard holdout slice, and a human signs off before production-wide rescoreing.

PROJECTS

- SREGym Fault Scenario Contribution**  | *Python, Kubernetes, Docker* 2026
- Reproduced Kubernetes [issue #128162](#) as a merged benchmark scenario: cumulative mutating-webhook timeouts exceed the API server's admission deadline and silently block all pod creation, with no crash to signal the fault.
 - Designed a four-property mitigation oracle ending in a fresh probe-pod admission test, which rejects reward-hacking fixes such as deleting all webhooks that mask the symptom without restoring the control-plane network path.
- Token-Efficient LLM Query Router**  | *AMD Hackathon Act II, Track 1* 2026
- Designed a four-tier inference cascade (regex router → ONNX classifier → LoRA fine-tuned Qwen2.5-1.5B → cloud LLM fallback) that resolves 8 task categories with minimal paid API usage, admitting each tier only at measured precision (router 64/64 on adversarial cases; sentiment classifier 100%).
 - Replaced model confidence checks with deterministic verification guards: sandboxed AST evaluation of model-emitted math expressions, which removes arithmetic errors and cuts output tokens ~90%, and an entity-completeness check that catches 99% of dropped entities in NER.
 - Fine-tuned the local tier on 2,800 programmatically validated synthetic examples and iterated 17 times against an LLM-as-judge harness mirroring the competition rubric, root-causing each regression before redeploy.
- KnowledgeRelay**  | *Python, FastAPI, LangChain, ChromaDB, React* 2025
- Built a knowledge-transfer agent that interviews outgoing team members with adaptive questions, ingests their documents and code, and answers new joiners' questions with source attribution. Retrieval rewrites follow-ups into self-contained queries.
- PrimeFaces Open Source Contribution**  | *Java, JSF, PrimeFaces* 2024
- Contributed a feature enhancement to PrimeFaces' `JPALazyDataModel` that lets developers inject custom filters into DataTable-backed JPA queries.

HONORS, AWARDS & CERTIFICATIONS

4th place out of 32 teams, Code Generation in Bangla Shared Task, BLP at IJCNLP-AAACL 2025 [\[Leaderboard\]](#)
6th place out of 30 teams, DSI AI Agent Hackathon 2025 [\[Leaderboard\]](#)
12th place in post-evaluation (19th official), Violence Inciting Text Detection, BLP at EMNLP 2023 [\[Leaderboard\]](#)
11th place, BUET CSE Fest 2022 AI Contest
Certified in Cybersecurity (CC), ISC2 [\[Badge\]](#)

TEACHING & SERVICE

- Course Instructor** 2024 – Present
Dynamic Learning *Remote*
- Authored and taught [Enterprise Web Development with JEE, JSF, and PrimeFaces](#): component-driven design, CDI/EJB integration, and JSF lifecycle management. 
- Volunteer**, IEEE International Conference on Bangla Speech and Language Processing (ICBSLP) 2019

TECHNICAL SKILLS

Languages: Java, Python, SQL, Bash, JavaScript, C/C++
Backend & Systems: Jakarta EE (JSF, CDI, EJB, JPA, JMS), Spring Boot, FastAPI, Kafka, GlassFish
AI & ML: PyTorch, LLMs, LoRA fine-tuning, RAG, LangChain, XGBoost
Data & Infrastructure: PostgreSQL, MySQL, Redis, MinIO, Docker, Kubernetes, Prometheus, Grafana, Loki, Nginx
Security: RBAC, JWT